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https://github.com/Kiru0310/Python-for-machine-learning/blob/main/ST-1b.pdf

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In []:

PROBLEM STATEMENT:
Write a python program to solve a system of linear equations represented by:
Ax=b
Where:
A is a 2*2 matrix
B is a 2*1 matrix
The program should compute and return the solution vector x

In [2]:

import numpy as np

def solve_linear_system(A_matrix, b_vector):
 """
 Solves the system Ax = b using NumPy.
 """
 try:
 # Convert inputs to numpy arrays
 A = np.array(A_matrix)
 b = np.array(b_vector)

 # Solve for x
 x = np.linalg.solve(A, b)
 return x

 except np.linalg.LinAlgError:
 return "Error: Matrix A is singular and cannot be solved."

A = [[3, 1],
 [1, 2]]

b = [9, 8]

solution = solve_linear_system(A, b)

print(f"Matrix A:\n{np.array(A)}")
print(f"Vector b: {b}")
print(f"Solution vector x: {solution}")

Matrix A:
[[3 1]
 [1 2]]
Vector b: [9, 8]
Solution vector x: [2. 3.]

In []:

